

Review: Taylor Series

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k$$

Taylor Formula

$$f(x) = \underbrace{\sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k}_{P_n(x)} + \underbrace{\frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}}_{R_n(x)} \quad \text{for some } c \in [a, x]$$

$P_n(x)$

$R_n(x)$: remainder or error term

TABLE 10.1 Frequently used Taylor series

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \cdots = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

$$\frac{1}{1+x} = 1 - x + x^2 - \cdots + (-x)^n + \cdots = \sum_{n=0}^{\infty} (-1)^n x^n, \quad |x| < 1$$

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad |x| < \infty$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad |x| < \infty$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad |x| < \infty$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n-1} \frac{x^n}{n} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}, \quad -1 < x \leq 1$$

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^n \frac{x^{2n+1}}{2n+1} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, \quad |x| \leq 1$$

The Binomial Series

For $-1 < x < 1$,

$$(1+x)^m = 1 + \sum_{k=1}^{\infty} \binom{m}{k} x^k,$$

where we define

$$\binom{m}{1} = m, \quad \binom{m}{2} = \frac{m(m-1)}{2!},$$

and

$$\binom{m}{k} = \frac{m(m-1)(m-2) \cdots (m-k+1)}{k!} \quad \text{for } k \geq 3.$$

2024 Calculus II Supplement Homework (4)

1. Exercise 42, (1),(3),(5),(7) and Exercise 43,(1) in page 28 from Calculus Supplement Homework.

2. Exercise 53 in page 29 from Calculus Supplement Homework.

3.(1). Express the function $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ as the Maclaurin series.

(2). Express the function $f(x) = x^2 - 5x + 6$ as the Taylor series at $x = 4$.

(3). Express the function $f(x) = \frac{1}{x^2+3x+2}$ as the Maclaurin series.

(4). Express the function $f(x) = \frac{1}{x^2-5x+6}$ as the Taylor series at $x = 4$.

4. Find the following limits.

(a) $\lim_{x \rightarrow 0} \left(\frac{1}{\ln(1+x)} - \frac{1}{e^x \sin x} \right).$ (b) $\lim_{n \rightarrow \infty} [(n^2 - n)e^{\frac{1}{n}} - \sqrt{n^4 + 1}].$

5 Let $f(x) = \tan x$. Find $f^{(5)}(0)$.

Q1, P29 Ex 42 (1)(3)(5)(7)

42. Find the sum function for the following series.

(1) $\sum_{n=1}^{\infty} (-1)^{n+1} n x^{n-1};$

(2) $\sum_{n=1}^{\infty} \frac{n(n+1)x^n}{(n-1)!};$

(3) $\sum_{n=1}^{\infty} (-1)^n \frac{1}{2n-1} \left(\frac{x-2}{3} \right)^{2n+1};$

(4) $\sum_{n=1}^{\infty} \frac{2n-1}{2^n} x^{2n-2};$

(5) $\sum_{n=1}^{\infty} n(n+1)x^n;$

(6) $\sum_{n=1}^{\infty} (-1)^{n-1} n^2 x^n;$

(7) $\sum_{n=0}^{\infty} \frac{n^2 + 1}{2^n n!} x^n;$

(8) $\sum_{n=1}^{\infty} (-1)^n \frac{n+1}{(2n+1)!} x^{2n+1}.$

Sol:

$$\begin{aligned}(1) \quad f(x) &= \sum_{n=1}^{\infty} (-1)^n x^{n-1} \\ \int f(x) dx &= - \sum_{n=1}^{\infty} x^n \frac{(-1)^n}{n} \\ &= - \frac{x}{1+x} \quad \text{或} \\ &= 1 - \frac{1}{x+1} \\ \therefore f(x) &= \frac{1}{(x+1)^2}\end{aligned}$$

$$\begin{aligned}g(x) &= \sum_{n=1}^{\infty} n x^{n-1} \\ \int g(x) dx &= \sum_{n=1}^{\infty} x^n = \frac{x}{1-x} \\ &= -1 + \frac{1}{1-x} \\ g(x) &= \frac{1}{(1-x)^2} \\ \therefore \sum_{n=1}^{\infty} (-1)^n n x^{n-1} &= g(-x) = \frac{1}{(1+x)^2}\end{aligned}$$

(3)

$$\begin{aligned}\text{Let } f(t) &= \sum_{n=1}^{\infty} (-1)^n \frac{1}{2n-1} t^{2n-1} \\ F(t) &= \frac{f(t)}{t^2} = \sum_{n=1}^{\infty} (-1)^n \frac{1}{2n-1} t^{2n-3} \\ F'(t) &= \sum_{n=1}^{\infty} (-1)^n t^{2n-2} = - \sum_{n=1}^{\infty} (-1)^{n-1} \cdot (t^2)^{n-1} = - \sum_{n=1}^{\infty} \frac{1}{1+t^2}\end{aligned}$$

$$\therefore F(t) = -\arctan t + C$$

$$\therefore F(0) = 0 \Rightarrow C = 0$$

$$\therefore F(t) = -\arctan t$$

$$\text{Then } f(t) = -t^2 \arctan t$$

$$\therefore \text{原式} = f\left(\frac{x-2}{3}\right) = -\left(\frac{x-2}{3}\right)^2 \arctan \frac{x-2}{3}$$

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(5)

$$\frac{f(x)}{x} = \sum_{n=1}^{\infty} n(n+1) x^{n-1}$$

$$\int \int \frac{f(x)}{x} dx dx = \sum_{n=1}^{\infty} x^{n+1} = \frac{x^2}{1-x} = \frac{x^2-1}{1-x} = -(x+1) - \frac{1}{x-1}$$

$$\frac{f(x)}{x} = \left(-\frac{1}{x-1} \right)'' = \left(\frac{1}{(x-1)^2} \right)' = \frac{-2}{(x-1)^3}$$

$$f(x) = \frac{-2x}{(x-1)^3}$$

(7) ① $g(x) = \sum_{n=0}^{\infty} \frac{(\frac{x}{2})^n}{n!} = e^{\frac{x}{2}}$

② $f(x) = \sum_{n=1}^{\infty} \frac{n(\frac{x}{2})^n}{(n-1)!} \Rightarrow F(x) = \sum_{n=1}^{\infty} \frac{n(\frac{x}{2})^{n-1}}{(n-1)!}$

$$\int F(x) dx = 2 \sum_{n=1}^{\infty} \frac{(\frac{x}{2})^n}{(n-1)!} = x \sum_{n=1}^{\infty} \frac{(\frac{x}{2})^{n-1}}{(n-1)!} = x e^{\frac{x}{2}}$$

$$\therefore F(x) = e^{\frac{x}{2}} \left(1 + \frac{x}{2} \right)$$

$$\Rightarrow f(x) = e^{\frac{x}{2}} \left(\frac{x}{2} + \frac{x^2}{4} \right)$$

$$\therefore f(x) + g(x) = e^{\frac{x}{2}} \left(1 + \frac{x}{2} + \frac{x^2}{4} \right)$$

P29 Ex43

43. Compute the following series.

$$(1) \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n(n+1)}{2^n};$$

Sol

$$\text{let } f(x) = \sum_{n=1}^{\infty} n(n+1)x^n$$

$$\text{By Ex43 (5)} \Rightarrow f(x) = \frac{2x}{(1-x)^3}$$

$$\therefore \text{原式} = -f\left(-\frac{1}{2}\right) = -\frac{-2 \cdot \frac{1}{2}}{\left(\frac{3}{2}\right)^3} = \frac{8}{27}$$

Q2 P29 Ex53

53. Find the Taylor series generated by $f(x) = \ln \frac{1+x}{1-x}$ at $x = 0$ and find an approximate value of $\ln 2$ to the error no greater than 10^{-4} .

Sol

$$f(x) = \ln \frac{1+x}{1-x} = \ln(1+x) - \ln(1-x) = \sum_{n=1}^{\infty} \left[\frac{(-1)^{n-1} x^n}{n} - \frac{(-1)^n (-x)^n}{n} \right]$$

$$= \sum_{n=1}^{\infty} \frac{(1+(-1)^{n-1}) x^n}{n}$$

估计 $\ln 2$: 令 $\frac{1+x}{1-x} = 2$, 解得 $x = \frac{1}{3}$.

error: $f^{(n+1)}(x) = \left(\frac{1}{1+x}\right)^{(n)} - \left(\frac{1}{x-1}\right)^{(n)} = n!(-1)^n [(x+1)^{-n-1} - (x-1)^{-n-1}]$

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} x^{n+1} \quad \text{where } c \in [0, x]$$

$$= \frac{n!(-1)^n [(c+1)^{-n-1} - (c-1)^{-n-1}]}{(n+1)!} x^{n+1}$$

$$= \frac{(-1)^n [(c+1)^{-n-1} - (c-1)^{-n-1}]}{n+1} x^{n+1}$$

$$|R_n(\frac{1}{3})| \leq \frac{[(0+1)^{-n-1} + (\frac{2}{3})^{-n-1}]}{n+1} \cdot (\frac{1}{3})^{n+1}$$

$$\leq \frac{\frac{1}{3^{n+1}} + \frac{1}{2^{n+1}}}{n+1}$$

$$\leq \frac{1}{2^n(n+1)} \quad \text{want: } < 10^{-4}$$

解得: $n \geq 10$

$$\therefore \ln 2 \approx \text{前 10 项和} \sum_{n=1}^{10} \frac{(1+(-1)^n)}{n} \cdot \frac{1}{3^n}$$

$$= 2 \cdot \left[1 \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3^3} + \frac{1}{5} \cdot \frac{1}{3^5} + \frac{1}{7} \cdot \frac{1}{3^7} + \frac{1}{9} \cdot \frac{1}{3^9} \right]$$

$$= 0.6931460474$$

Q3

3.(1). Express the function $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ as the Maclaurin series.

(2). Express the function $f(x) = x^2 - 5x + 6$ as the Taylor series at $x = 4$.

(3). Express the function $f(x) = \frac{1}{x^2 + 3x + 2}$ as the Maclaurin series.

(4). Express the function $f(x) = \frac{1}{x^2 - 5x + 6}$ as the Taylor series at $x = 4$.

Sol:

$$(1) f(x) = a_0 + a_1 x + \dots + a_n x^n$$

$$\begin{aligned} (2) f(x) &= x^2 - 5x + 6 \\ &= x^2 - 8x + 16 + 3x - 10 \\ &= (x-4)^2 + 3x - 12 + 2 \\ &= 2 + 3(x-4) + (x-4)^2 \end{aligned}$$

$$\begin{aligned} (3) f(x) &= \frac{1}{x^2 + 3x + 2} = \frac{1}{x+1} - \frac{1}{x+2} = \frac{1}{1-(-x)} - \frac{1}{2} \frac{1}{1-(-\frac{1}{2}x)} \\ &= \sum_{n=0}^{\infty} (-x)^n - \frac{1}{2} \sum_{n=0}^{\infty} \left(-\frac{x}{2}\right)^n \\ &= \sum_{n=0}^{\infty} \left[(-1)^n + \left(-\frac{1}{2}\right)^{n+1}\right] x^n \end{aligned}$$

$$\begin{aligned} (4) f(x) &= \frac{1}{x^2 - 5x + 6} = \frac{1}{x-3} - \frac{1}{x-2} = \frac{1}{1-[-(x-4)]} - \frac{1}{2} \frac{1}{1-\left[\frac{-(x-4)}{2}\right]} \\ &= \sum_{n=0}^{\infty} [-(x-4)]^n - \frac{1}{2} \sum_{n=0}^{\infty} \left[\frac{-(x-4)}{2}\right]^n \\ &= \sum_{n=0}^{\infty} \left[(-1)^n + \left(-\frac{1}{2}\right)^{n+1}\right] (x-4)^n \end{aligned}$$

□

4. Find the following limits.

(a) $\lim_{x \rightarrow 0} \left(\frac{1}{\ln(1+x)} - \frac{1}{e^x \sin x} \right)$. (b) $\lim_{n \rightarrow \infty} \left[(n^2 - n)e^{\frac{1}{n}} - \sqrt{n^4 + 1} \right]$.

Sol.

(a)

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{e^{x \sin x} - \ln(1+x)}{e^x \ln(1+x) \cdot \sin x} \\ &= \lim_{x \rightarrow 0} \frac{(1+x)X - \left(X - \frac{X^2}{2}\right)}{e^x X^2} \\ &= \lim_{x \rightarrow 0} \frac{\frac{3}{2}X^2}{e^x X^2} \\ &= \frac{3}{2} \end{aligned}$$

Taylor series.

(b)

$$\begin{aligned} & \lim_{n \rightarrow \infty} \frac{\left(1 - \frac{1}{n}\right)e^{\frac{1}{n}} - \sqrt{1 + \frac{1}{n^4}}}{\frac{1}{n^2}} \\ &= \lim_{x \rightarrow 0} \frac{(1-x)e^x - \sqrt{1+x^4}}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{(1-x)\left(1+x+\frac{x^2}{2}\right) - \left(1+\frac{1}{2}x^4\right)}{x^2} \\ &= -\frac{1}{2} \end{aligned}$$

5 Let $f(x) = \tan x$. Find $f^{(5)}(0)$.

5. Let $f(x) = \tan x$. Find $f^{(7)}(0)$.

Solution:

Suppose $\tan x = a_0 + a_1x + \frac{a_2}{2!}x^2 + \dots + \frac{a_n}{n!}x^n + \dots$, where $a_n = f^{(n)}(0)$.

Since f is an odd function, $f^{(2k-1)}$ is an even function. Thus, $a_{2k} = f^{(2k)}(0) = 0$.

Using the formula $(\tan x)(\cos x) = \sin x$, we have

$$\begin{aligned} & \left(a_1x + \frac{a_2}{3!}x^3 + \frac{a_4}{5!}x^5 + \frac{a_6}{7!}x^7 + \dots \right) \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \right) \\ &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \end{aligned}$$

Hence, $a_1 = 1$. And, it follows that

$$-\frac{1}{2} + \frac{a_3}{3!} = -\frac{1}{3!}. \quad \text{So, } a_3 = 2.$$

$$\frac{1}{4!} - \frac{1}{6} + \frac{a_5}{5!} = \frac{1}{5!}. \quad \text{So, } a_5 = 16.$$

$$-\frac{1}{6!} + \frac{1}{72} - \frac{1}{16} + \frac{a_7}{7!} = -\frac{1}{7!}. \quad \text{So, } f^{(7)}(0) = a_7 = 272.$$